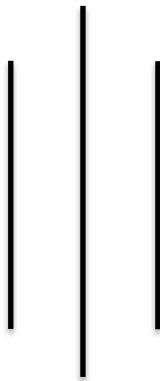
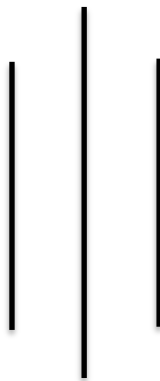


Field Visit Report on Bagmati Irrigation Project, Sarlahi



Submitted to:

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I would also like to extend my profound appreciation to the **Bagmati Irrigation Project team**, for their invaluable support and cooperation during our visit. Their expertise, professionalism, and willingness to share in depth technical knowledge greatly enriched our understanding of the project's planning, implementation, and operational challenges.

I am also grateful to my classmates for their cooperation and active participation, making this visit both educational and enjoyable. Lastly, my sincere appreciation to everyone who contributed, directly or indirectly, to the success of this field visit.

PREFACE

This report is prepared after the successful conduction of a field visit under the department of Civil Engineering of Pashchimanchal Campus to Karmaiya, Sarlahi, according to the curriculum of Irrigation Engineering on 30 January 2025. I studied, observed and analyzed various aspects of irrigation engineering, including water resource management, canal systems, and irrigation structures.

The Bagmati Irrigation Project is a significant initiative that utilizes the Bagmati River to support agricultural irrigation. It consists of a well-planned network of canals, diversion structures, and cross-drainage works, ensuring efficient water distribution to farmlands. The project plays a crucial role in enhancing agricultural productivity by providing a reliable water supply, particularly during dry seasons.

This report presents a comprehensive overview of the field visit, including technical observations, key learnings, and practical applications related to irrigation engineering. It also discusses the challenges associated with the project and highlights insights gained from interactions with project officials. I hope this document serves as a valuable reference for future studies in irrigation engineering.

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1. INTRODUCTION

Irrigation is a fundamental component of Nepal's agricultural sector, where over 60% of the population depends on farming for their livelihood. Given the country's diverse topography and uneven water distribution, efficient irrigation systems are essential to ensure year-round agricultural productivity. Nepal's irrigation infrastructure primarily relies on surface water from rivers, supplemented by groundwater extraction in some regions. Out of the total 14.718-million-hectare area of the country only 2.641 million hectare is arable, and 1.766 million hectare is irrigable. 76% of potential irrigable area lies in Terai region of Nepal where there is low rainfall.

The Bagmati Irrigation Project (BIP), located in Sarlahi district, is one of the major irrigation initiatives in the Terai region, designed to enhance agricultural output by providing a regulated and reliable water supply to farmlands. The project diverts water from the Bagmati River through an extensive network of canals, benefiting thousands of hectares of cultivated land.

This field visit was conducted as part of the civil engineering curriculum to provide us with practical exposure to irrigation infrastructure, water distribution techniques, and management strategies. The study focused on the project's diversion weir, main canal, branch canals, water regulation structures, and maintenance practices. Bagmati Irrigation Project provided an in-depth understanding of irrigation planning, canal hydraulics, and water resource management in Nepal's context. This visit also helped bridge the gap between theoretical concepts and field applications, reinforcing key principles from irrigation engineering.

This report presents a comprehensive analysis of the field visit, covering the technical aspects of the Bagmati Irrigation Project, key learnings from visit, and the challenges associated with its implementation and maintenance.

2. OBJECTIVES

1. To study the diversion headwork and its components from intake to the canal outlet.
2. To examine regulating structures, cross-drainage works, and major water sources.
3. To assess the command area and canal design discharge.

3. METHODOLOGY

1. Preliminary Study and Literature Review
 - Reviewed relevant textbooks and gathered background information on the Bagmati Irrigation Project.
2. On-Site Observation and Data Collection
 - Observed and recorded data on canals, diversion structures, and water flow characteristics.
3. Technical Discussions and Expert Interaction
 - Engaged with experts to discuss operational challenges and irrigation practices.
4. Photographic and Visual Documentation
 - Captured photographs and sketches of key structures for analysis and reference.
5. Data Analysis and Report Compilation
 - Analyzed field data and compiled findings into a report, correlating observations with theoretical principles.

4. OVERVIEW OF BAGMATI IRRIGATION PROJECT

4.1 Project Background and History

The Bagmati Irrigation Project (BIP) is one of Nepal's national priority irrigation projects, situated in the Sarlahi district and draws water from the Bagmati River, a perennial river originating from the Himalayan region. The project's catchment area covers approximately 2,720 square kilometers, and its command area extends across multiple districts, ensuring year-round irrigation to extensive agricultural lands.

The Bagmati Irrigation Project was initiated in the fiscal year 1978/79 with the primary objective of diverting water from the Bagmati River for irrigation purposes. In its first phase, the project aimed to irrigate 37,000 hectares, comprising 14,000 hectares in Sarlahi district and 23,000 hectares in Rautahat district. The project's second phase envisions an expansion to 122,000 hectares, covering the Sarlahi, Rautahat, Bara, Mahottari, and Dhanusha districts.

The project consists of major structural components that facilitate efficient water diversion and distribution across the command area.

Barrage: The Bagmati River Barrage is a 403.5-meter-long structure located upstream of the East-West Highway. It serves as the primary diversion structure with 30 no. of gates.

Eastern Main Canal: This canal extends 17.1 kilometers, with a discharge capacity of 64.4 cubic meters per second (cumecs). It further branches into three major distributaries:

- Manpur Branch Canal
- Gadhaiya Branch Canal
- Laxmipur Branch Canal

These canals collectively irrigate 14,000 hectares of farmland.

Eastern Head Regulator width = 37m, no. of gates = 7

Western Main Canal: Extending 28.2 kilometers, this canal supplies water with a discharge capacity of 48.3 cubic meters per second to an additional 23,000 hectares of agricultural land through three branch canals:

- Bhalohiya Branch Canal
- Bahuyari Branch Canal
- Baghi Branch Canal

Western Head Regulator width = 26m, no. of gates = 5

The Bagmati River's water level drops to 8-9 cumecs in the dry season, making it difficult to meet irrigation demands for the two canals. To address this, the Government of Nepal conceptualized an alternative water diversion approach, **Sunkoshi Marine Diversion project** in 1935, aiming to enhance water supply during low flow periods and support agricultural irrigation.

- **Sunkoshi-Marin Diversion Multipurpose Project (SMDP):** This initiative involves the diversion of water from the Sunkoshi River to the Marin River, ultimately supplementing the Bagmati Irrigation Project.
- **Project Components:**
 - A 13.3-kilometer tunnel will transfer a constant discharge of 67 cubic meters per second from Sunkoshi to Bagmati.
 - The diverted water will facilitate year-round irrigation to 122,000 hectares of farmland.
 - The project includes a hydropower component designed to generate 31.07 megawatts of electricity.

As of May 2024, a significant milestone was achieved with the breakthrough of the diversion tunnel, marking substantial progress toward project completion.



Fig.1- Details of Bagmati Irrigation Project

4.2 Location

The project area lies between longitudes $85^{\circ}17'$ to $85^{\circ}36'$ East and latitude $26^{\circ}46'$ to $27^{\circ}6'$ North and Comprises of land in the Sarlahi district of Janakpur Zone and Rautahat district of Narayani Zone of the Central Development Region. The command area of the project lies in the south of the East-West highway and north is the Churia hills. It is a flat plain with ground elevations varying typically, between 60m above mean sea level on the southern border with India to 130m at the boundary of Siwalik and is called the Terai. The major rivers within the project area are Lal Bakeya, Lamaha, Chandi, Bagmati, Lakhandei and Jhim. The source of water for irrigation supply is the Bagmati River.

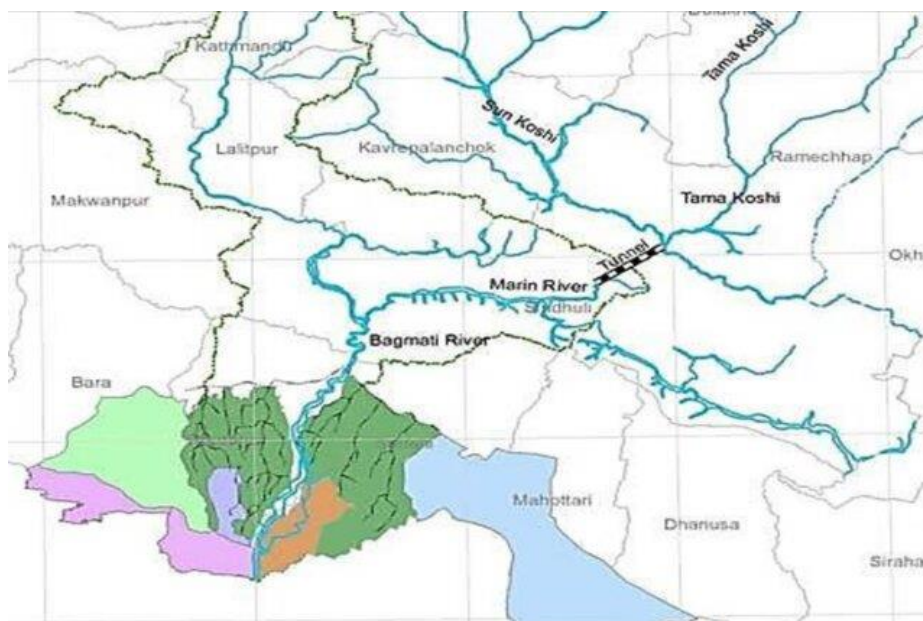


Fig. 2- Project Area (src: researchgate.net © Mukesh kifle)

4.3 Information and Insights

Design Flood: 100-year return period

Highest flood recorded : 4th and 5th Shrawan 2050 (20th july 1993) , flood level 134.40m.

Construction Cost: NPR 37,809,610.25 (estimated)

Maintenance Cost: 1.5 to 2 crores yearly

Owner: Ministry of Water Resources, Department of Irrigation, Government of Nepal

Financier: Government of Nepal, Asian Development Bank (ADB)

Sedimentation Problem: Siltation problem is high due to deforestation and mobile boundary.

Provision in case of High Flooding:

- i. The system can withstand 10000 cumecs of water.
- ii. Hydraulic and Structural Design is adopted.

River Training Work Adopted: Guide Banks, Marginal Bunds, Pitched Island, Bandalling

Regulating Structure provided: Head Regulators

Cross Drainage Structure: Syphon Super passage, Aqueduct

Major Causes of Water Losses:

- i. Seepage losses
- ii. High Percolation
- iii. Not closing the gate properly

Major Crops in Command Area: Paddy, Rahar Dal

Future Project: Sunkoshi-Marine Diversion Project

Mr. Binaya Kumar Gami, Senior Divisional Engineer of the Bagmati Irrigation Project, emphasized that the project has managed to irrigate only 45,600 hectares in the Sarlahi and Rautahat districts due to the government's lack of focus on this important national initiative.

5. OBSERVATIONS AND ANALYSIS

We collected and analyzed different information to meet our objective of field visit of irrigation project.

Diversion Headwork is the hydraulic structure constructed across a river for the purpose of raising water level in the river so that it can be diverted to the off-taking canal. Its function is to raise the water level on the upstream side, control the entry of silt into canals and regulate the intake of water to the canal.



Fig.3- Diversion Headwork of Bagmati Irrigation Project

Various components from the headwork to the canal outlet were observed.

- a) Barrage
- b) Under-Sluices
- c) Divide wall
- d) Canal Head Regulator
- e) Canal Portion
- f) Fish Ladder
- g) Silt Ejectors
- h) Weir's Ancillary works such as shutters, gates, etc
- i) Cross-drainage structures
- j) By pass Canal

5.1 Barrage

A barrage is constructed across the Bagmati River to regulate the flow of water and ensure a controlled intake into the irrigation canal. A raised concrete crest wall where ponding of water is achieved by shutters or gates. The barrage has a design discharge of $8000\text{m}^3/\text{s}$ and has a bypass channel of $2500\text{m}^3/\text{s}$ discharge. Therefore, the barrage can handle a discharge of $10500\text{m}^3/\text{s}$.

5.2 Under-Sluices

The under sluices are the openings provided at the base of the barrage. These openings are provided with adjustable gates. Normally, the gates are kept closed. The crest of the under-sluice portion of the weir is kept at the lower level (1-1.5m) than the crest of normal portion of the weir. Its functions are:

1. **Silt Control** – Regulates sediment entry and prevents silt deposition near the canal head.
2. **Flow Regulation** – Maintains water levels and controls discharge into the canal.
3. **Flood Management** – Helps pass excess water and reduces flood risks.
4. **Structural Protection** – Prevents damage to the barrage and canal headworks.



Fig. 4 - Under-sluice

5.3 Divide Wall

The divide wall is a masonry or concrete wall constructed at the right angle to the axis of the weir. The divide wall extends on the upstream side beyond the beginning of canal head regulator, and on the downstream side, it extends up to the ends of loose protection of undersluices. In this irrigation project, cement concrete divide wall was observed.



Fig. 5 - Divide Wall

5.4 Canal Head Regulator

A structure which is constructed at the head of the canal to regulate flow of water is known as canal head regulator. It controls the amount, velocity, and sediment of water into the canal system. The project includes canal head regulators situated on both the eastern and western sides of the barrage. Components of a Canal Head Regulator are:

1. **Gates/Shutters** – Adjustable gates that regulate water entry.
2. **Silt Excluders** – Structures designed to minimize sediment entry.
3. **Guide Walls & Divide Walls** – Directs flow and separates different canal sections.
4. **Abutments & Piers** – Provides structural stability to the regulator.

5.5 Canal Portion

The main canal was lined with cement concrete and is in working condition. The distributary canal was not working, and the components were on the deteriorated state.



Fig. 6 - Main Canal

5.6 Fish Ladder

A fish ladder (also called a fishway or fish pass) is a structure built alongside barrages, weirs, or dams in irrigation projects to allow fish to migrate upstream or downstream despite water barriers. The Bagmati Irrigation Barrage is equipped with two fish ladders, each measuring 1.5 meters in width and 4 meters in length. The ladders are constructed to create a series of steps or pools, enabling fish to navigate upstream and downstream past the barrage.

5.7 Silt Ejectors

Silt ejectors have been implemented in its western main canal system to manage sedimentation and maintain efficient water flow. Silt ejectors are hydraulic structures designed to remove excess silt from canals, preventing sediment accumulation that can hinder water delivery and reduce the canal's capacity.

5.8 Cross Drainage Structures

Cross-drainage structures in the Bagmati Irrigation Project (BIP) are designed to manage the interaction between irrigation canals and natural drainage systems. These structures help maintain smooth water flow while preventing flooding and erosion.

Key Cross-Drainage Structures in BIP:

1. Level Crossing: Observed in western channel.

- Allow canal water to flow over natural drainage channels at the same level, controlled by gates or weirs.
- During floods, gates open to prevent canal flooding.

2. Inverted Siphon:

- Enables canal water to pass beneath rivers or drains, maintaining irrigation flow without interrupting natural water courses.

3. Super Passage:

- This structure allows a canal to pass over a natural drain or river, ensuring uninterrupted flow.

4. Aqueduct:

- Used to convey canal water across natural drainage paths, ensuring the canal remains functional even when intersecting with depressions or streams.

These structures prevent damage from floods, erosion, and sedimentation. They ensure the efficient and continuous flow of water through the canal system for irrigation.

5.9 Bypass Canal

The Bagmati Irrigation Project (BIP) includes bypass canals as part of its flood management and sediment control system. These canals help in handling excess water from the Bagmati River, especially during the monsoon season, preventing flooding and sediment accumulation in the main canal system.



Fig. 7 - Bypass Canal

6. CHALLENGES IDENTIFIED

6.1 Technical Challenges

The irrigation system faces issues such as sedimentation, seepage, and infrastructure deterioration, reducing its efficiency and increasing maintenance needs. Additionally, the lack of automation in flow regulation leads to delays and uneven water distribution.

6.2. Operational Challenges

Unequal water distribution, poor maintenance management, and limited farmer involvement hinder the system's long-term sustainability. Insufficient financial and technical resources further delay essential repairs and improvements.

6.3. Environmental Challenges

Seasonal water scarcity, encroachment, and climate variability significantly impact irrigation reliability. Unpredictable river flow and human activities near canals further threaten the system's stability.

7. CONCLUSION AND DISCUSSIONS

Our field visit to the Bagmati Irrigation Project was successfully conducted, meeting all our predefined objectives. The visit provided valuable insights into the practical aspects of irrigation infrastructure and its significance in Nepal's agricultural sector. As a country where nearly 80% of the population depends on agriculture, effective irrigation management is essential for economic growth and sustainability.

Through our observations, we concluded that large-scale irrigation projects should be operated with systematic management and regular maintenance to ensure efficiency and longevity. The Nepalese government, along with local authorities, should periodically assess these projects to optimize their performance and economic impact. The Bagmati Irrigation Project plays a crucial role in sustaining agricultural productivity, contributing both directly and indirectly to the livelihoods of farmers. However, we identified several operational challenges, including sediment deposition in canals, occasional blockages, and silt accumulation, all of which hinder the project's effectiveness.

Addressing these issues is crucial to maintaining a steady water supply and ensuring the project functions optimally. The committee responsible must remain proactive in identifying and resolving these concerns. Overall, the field visit was a valuable learning experience, allowing us to practically comprehend the theories studied in our classrooms.

8. RECOMMENDATIONS

To ensure the long-term efficiency and sustainability of the Bagmati Irrigation Project, the following measures should be implemented:

- Conduct regular maintenance and monitoring to prevent sediment accumulation and structural issues.
- Implement effective desilting mechanisms to ensure continuous water flow.
- Upgrade outdated components with durable materials to enhance efficiency and longevity.
- Integrate advanced irrigation technologies like automated flow control and real-time monitoring systems.
- Provide technical training to personnel for efficient system management and troubleshooting.
- Engage local farmers and stakeholders in decision-making and maintenance processes.
- Encourage government and private sector investments to expand the project's scope and sustainability.
- Develop contingency plans to mitigate risks from environmental challenges like floods and droughts.

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